Example

Minimize $0.5x^2 + 5x + 4y$

Minimize $0.5x^2 + 0y^2 + 0xy + 5x + 4y$

Minimize $\frac{1}{2}[x^2 + 0y^2 + 0xy] + 5x + 4y$

$\frac{1}{2}x^T Px + q^T x$

$P = [0\ 0]$ $\quad q = [4]$

Subject to

$x \geq 0$

$y \geq 0$

$x + 5y \leq 15$

$2x + 5y \leq 100$

$3x + 4y \leq 80$

$\rightarrow$

$-x \leq 0$

$-y \leq 0$

$-x - 5y \leq -15$

$2x + 5y \leq 100$

$3x + 4y \leq 80$

$Gx \leq h$

$$G = \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ -1 & -3 \\ 2 & 5 \\ 3 & 4 \end{bmatrix} \quad h = \begin{bmatrix} 0 \\ 0 \\ -15 \\ 100 \\ 80 \end{bmatrix}$$

x=0, y=5

$x + 5y - 15 = 0$

$2x + 5y - 25 = 0$

$3x + 4y - 20 = 0$

$x \geq 0$

$y \geq 0$

Minimum value:

$0.5x^2 + 5x + 4y = 20$